

Mahendran M

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Education

Gouvernement College of Engineering

Bachelor of Engineering , Computer Science (GPA:7.4/10.00)

May 2021

Theni,TN, INDIA

Experience

SBNA Software Solutions

Frontend Engineer

Aug 2021 – Present

Coimbatore,TN, INDIA

- Collaborated with cross-functional teams (Product, QA, Backend, DevOps) to refine requirements, resolve UI/UX issues, and deliver production-ready React applications
- Engineered scalable React.js frontends using Hooks, TypeScript, Redux Toolkit, Zustand, and React Query, improving UI responsiveness and maintainability by 30%.
- Built an AI-powered Agent Contact Center interface with multi-channel chat, voice, and video using sip.js + FreeSWITCH, enabling seamless handling of 1,000+ concurrent agent interactions.
- Integrated Azure AI Agent to support automated knowledge search, intent prediction, and workflow guidance, reducing manual resolution time by 35%.
- Developed backend APIs using Node.js and Express.js to support AI workflows, form automation, and contact center features, ensuring reliable data exchange between frontend and backend systems.
- Implemented secure REST endpoints, middleware, and authentication flows (JWT/Firebase Auth), improving application security and stability for React and React Native clients.
- Optimized frontend performance through memoization, virtualization, lazy loading, and code splitting, reducing unnecessary re-renders by 25% and improving application reliability.

Projects

Agent Contact Center | React.js, Redux, Typescript, SIP.js, Node JS

- Built an AI-powered React contact center UI with multi-channel chat/call support to streamline agent workflows.
- Optimized the Agent Widget by restructuring components and state updates, improving performance by 30%.
- Customized sip.js with WebSocket reconnection and call-recovery logic to improve reliability on unstable networks
- Developed a drag-and-drop IVR Flow Builder enabling non-technical teams to design call flows 60% faster.

245D- Form Filling Guide | React.js, Typescript, Azure AI Agent, Azure Cognitive Search, Azure Open AI, Node JS

- Built an AI-assisted form completion guide with real-time validation, error detection, and field recommendations using Azure Cognitive Search and LLM reasoning
- Reduced form completion time by 40% through automated guidance and contextual suggestions.

Denial management - AI Agent | React.js, Typescript, Azure AI Agent, Azure Cognitive Search, Node JS, Azure OpenAPI

- Developed a claims assistance chatbot using Azure AI Agent to automate claim status queries, document parsing, and policy lookup.
- Implemented Cognitive Search for fast knowledge retrieval, reducing manual customer support workload by 35%

Technical Skills

Languages: JavaScript (ES6+), TypeScript, HTML5, CSS3

Frontend: React.js, Redux Toolkit, React Query, TailwindCSS, React Native

Backend: Node.js, Express.js, REST API Development, Authentication & Authorization (JWT, Firebase Auth)

Database: MongoDB, Mongoose, Schema Design, Aggregation Pipelines, PostgreSQL

Real-Time & Integrations: WebSockets, SIP.js, WebRTC, Azure AI Agent, Azure Cognitive Search

DevOps & Tools: Docker, Nginx, Git, GitHub, CI/CD, Claude Code

Testing: Jest, React Testing Library

Concepts: Component Architecture, Performance Optimization, State Management, Real-Time Communication